Project Report

Ethnic Bias and Generalization in Facial Attractiveness Prediction

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Abstract

[Abstract text here - max 200 words]

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1. Introduction

1.1. Problem Statement

Automated systems for rating or ranking human faces are increasingly used in applications ranging from online platforms to hiring tools. Unlike many computer vision tasks, facial attractiveness prediction is inherently subjective and culturally shaped, making it particularly prone to encoding demographic bias. Models trained on data from a narrow set of ethnic groups may learn cultural beauty standards rather than broadly applicable visual patterns, raising both methodological and ethical concerns.

1.2. Research Questions

Our core research questions are:

1. How well do models generalize across datasets with different ethnic compositions, and does training on more ethnically diverse data reduce per-group performance disparities?
2. Which geometric facial measurements drive attractiveness predictions across ethnic groups, and how well do interpretable landmark-based features perform?
3. To what extent does the choice of evaluation methodology (stratified vs. plain cross-validation) affect reported model performance and fairness metrics?

1.3. Scope and Contributions

We formulate facial attractiveness prediction as a supervised regression task where the goal is to predict the mean attractiveness rating assigned by human annotators to a face image. Our main contributions are:

- Combination and harmonization of three publicly available datasets differing in ethnic composition and scale
- Extraction of 39 interpretable geometric and expression features from MediaPipe Face Mesh landmarks
- Comprehensive evaluation of 10 regression models with stratified cross-validation and held-out test set

- Per-ethnicity fairness analysis quantifying performance disparities
- Cross-dataset generalization analysis revealing dataset-specific bias
- SHAP-based feature importance analysis identifying which facial measurements drive predictions

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[Dataset overview]

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[Top features: Eye width ratio, Nose width ratio, Mouth-chin distance, Gonial angle, Jaw width ratio]

Model	MAE	RMSE	Pearson r
MLP	**0.5438**	**0.7106**	**0.7143**
Ensemble (Stacking)	0.5543	0.7148	0.7098
CatBoost	0.5555	0.7156	0.7092
XGBoost	0.5540	0.7169	0.7080

Table 5.1.: Top Model Performance on Held-Out Test Set

Ethnicity	**N**	**MAE**	**Pearson r**	**Delta**
Asian	2,871	0.5217	0.7425	—
Black	59	0.5842	0.5807	+12.0%
Caucasian	496	0.5979	0.5932	+14.6%
Hispanic	59	0.7333	0.4995	+40.6%
Indian	31	1.0273	0.3072	+96.9%
Middle Eastern	58	0.6844	0.6444	+31.2%

Table 5.2.: Per-Ethnicity Fairness (MLP Best Model)

Train	**Test**	**MAE**	**r**
LiveBeauty	MEBeauty	0.8566	0.4178
MEBeauty	SCUT	0.7700	0.3610
SCUT	MEBeauty	0.8487	0.2800
Combined (5-fold CV)	**0.5683**	**0.6803**	

Table 5.3.: Cross-Dataset Generalization Performance

6. Discussion

[Discussion of results]

7. Conclusions

[Conclusions]

Bibliography

A. Additional Results

[Additional tables/figures]

Ehrenwörtliche Erklärung

Ich versichere, dass ich die beiliegende Bachelor-, Master-, Seminar-, oder Projektarbeit ohne Hilfe Dritter und ohne Benutzung anderer als der angegebenen Quellen und in der untenstehenden Tabelle angegebenen Hilfsmittel angefertigt und die den benutzten Quellen wörtlich oder inhaltlich entnommenen Stellen als solche kenntlich gemacht habe. Diese Arbeit hat in gleicher oder ähnlicher Form noch keiner Prüfungsbehörde vorgelegen.

Declaration of Used AI Tools

Tool	Purpose	Where?	Useful?
Claude	chapter declaration		+

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